Part D: LLM Prediction System — RAG-Based AI Buyer Agent

MGMT 687 – AI for Business Decisions | Team Project

Purpose: Deploy a multimodal LLM as an AI buyer agent that directly evaluates and ranks the 5 new products. Three prompting strategies are compared: Zero-Shot, Few-Shot (random), and RAG-Enhanced Few-Shot (retrieves the most similar training products per query).

RAG Rationale: Instead of showing the LLM 4 random examples, each test product is paired with its *k* most similar training products (by cosine similarity on pre-computed title embeddings). This means the LLM effectively learns from the full training set — retrieving the most relevant subset for each query. Pre-computed embeddings from `features_train.csv` are reused to avoid re-running the ~14-minute Part B extraction pipeline.

Inputs: `dataset_history.csv`, `dataset_new.csv`, `features_train.csv` (embeddings only) **Outputs:** LLM priority ranking, strategy comparison charts

```
# ── Install Dependencies ──────────────────────────────────────────
!pip install groq pillow requests pandas numpy matplotlib seaborn scikit-learn
```

```
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```

```
# ── Imports ───────────────────────────────────────────────────────
import pandas as pd
import numpy as np
import requests
import base64
import json
import time
import re
import warnings
import matplotlib.pyplot as plt
import seaborn as sns
from io import BytesIO
from PIL import Image
from groq import Groq
from sklearn.metrics.pairwise import cosine_similarity
```

```
warnings.filterwarnings('ignore')
plt.rcParams.update({'figure.dpi': 130, 'font.size': 11})
np.random.seed(42)
print('Libraries loaded.')
```

```
Libraries loaded.
```

Step 1: Load Data

> **Note on `features_train.csv`:** Only the `title_embedding_vector` column is used from this file — solely to enable cosine-similarity retrieval for RAG-based few-shot selection. This avoids re-running the ~14-minute Groq extraction pipeline from Part B. No LLM-extracted features or ML model outputs are used in Part D. The LLM buyer agent evaluates each product independently from raw image and title alone.

```
# ── Upload and Load ──────────────────────────────────────────────────
from google.colab import files
print('Upload: dataset_history.csv, dataset_new.csv, features_train.csv')
uploaded = files.upload()

df_history  = pd.read_csv('dataset_history.csv')   # raw training data
df_test     = pd.read_csv('dataset_new.csv')        # 5 products to rank
df_features = pd.read_csv('features_train.csv')     # for embeddings only

# Keep only what we need from history
df_history = df_history[['title','image','price','ordered']].copy()
df_test    = df_test.reset_index(drop=True)

print(f'History : {df_history.shape}  |  Ordered: {df_history["ordered"].sum()}
print(f'Test    : {df_test.shape}')
print(f'\nTest products to rank:')
for i, row in df_test.iterrows():
    print(f'  [{i+1}] {row["title"][:70]}  |  ${row["price"]:.2f}')
```

```
Upload: dataset_history.csv, dataset_new.csv, features_train.csv
```

Choose files 3 files

features_train.csv(text/csv) - 13268149 bytes, last modified: 06/03/2026 - 100% done
dataset_new.csv(text/csv) - 937 bytes, last modified: 05/03/2026 - 100% done
dataset_history.csv(text/csv) - 61723 bytes, last modified: 05/03/2026 - 100% done

```
Saving features_train.csv to features_train.csv
Saving dataset_new.csv to dataset_new.csv
Saving dataset_history.csv to dataset_history.csv
History : (331, 4)  |  Ordered: 105 / 331
Test    : (5, 4)

Test products to rank:
  [1] Men's Vintage Premium Washed Shirt Navy Blue / L  |  $59.09
  [2] Ribbed Textured Knit Polo Shirt (US Only) Khaki / L  |  $35.67
  [3] Men's Rose Print Vintage Casual Button Short Sleeve Shirt WHITE / L  |  $2
  [4] Classic Casual Button Down Cotton Linen Shirt (Us Only) Red / L  |  $35.96
  [5] Men's Casual Geo Pattern Short Sleeve Shirt & Shorts 2Pcs Set Burgundy  |
```

Step 2: Parse Training Embeddings for RAG Retrieval

```
# ── Parse Title Embeddings from features_train.csv ──────────────────
# Only title_embedding_vector is used — 384-dim from all-MiniLM-L6-v2 (Part
def parse_embedding_col(series, dim):
    vecs = []
    for val in series:
        try:
            v = json.loads(str(val))
            vecs.append(v if len(v) == dim else [0.0]*dim)
        except Exception:
            vecs.append([0.0]*dim)
    return np.array(vecs, dtype=np.float32)

TITLE_DIM = 384
print('Parsing title embeddings from training features...')
train_title_vecs = parse_embedding_col(df_features['title_embedding_vector']
print(f'Training embeddings shape: {train_title_vecs.shape}')

# Attach to history df for retrieval
df_history = df_history.iloc[:len(train_title_vecs)].copy().reset_index(dro
print(f'Training products available for retrieval: {len(df_history)}')
```

```
Parsing title embeddings from training features...
Training embeddings shape: (331, 384)
Training products available for retrieval: 331
```

Step 3: Groq API Setup

```
# ── API Configuration ──────────────────────────────────────────────────
GROQ_API_KEY  = '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'    #
VISION_MODEL  = 'meta-llama/llama-4-scout-17b-16e-instruct'
REQUEST_DELAY = 2.5

client = Groq(api_key=GROQ_API_KEY)

try:
    _t = client.chat.completions.create(
        model=VISION_MODEL,
        messages=[{'role':'user','content':'Reply: OK'}],
        max_tokens=5
    )
    print('API: ✓', _t.choices[0].message.content.strip())
except Exception as e:
    print(f'API error: {e}')

AGENT_SYSTEM_PROMPT = (
    'You are an experienced online fashion buyer browsing a product catalogue.
    'You evaluate each product purely on its commercial appeal and likelihood c
    'attracting purchase orders from real consumers browsing a typical e-commer
    'You always respond ONLY with a valid JSON object — no markdown, no extra t
)

def fetch_b64(url, timeout=8):
    try:
        r = requests.get(url, timeout=timeout, headers={'User-Agent': 'Mozilla/
        if r.status_code == 200:
            img = Image.open(BytesIO(r.content)).convert('RGB')
            buf = BytesIO(); img.save(buf, format='JPEG')
            return base64.b64encode(buf.getvalue()).decode(), 'image/jpeg'
    except Exception:
        pass
    return None, None

print('Agent ready.')
```

```
API: ✓ OK
Agent ready.
```

Step 4: Zero-Shot Prediction

No examples provided. The LLM relies entirely on its pre-trained world knowledge about fashion e-commerce and consumer behaviour.

```
# ── Zero-Shot Prompt and Prediction ──────────────────────────────
ZS_PROMPT_TEMPLATE = '''You are an online fashion buyer evaluating a product li
Product Title: "{title}"
Price: ${price:.2f}

Based on the product image and title, estimate the probability (0.0 to 1.0) tha
online shopper will purchase this product under standard advertising exposure.

Respond ONLY with this JSON:
{{
  "purchase_probability": <float 0.0-1.0>,
  "confidence": <"low"|"medium"|"high">,
  "key_reason": <one concise sentence>
}}'''

def call_llm(system_prompt, content, max_tokens=250, max_retries=3):
    """Generic LLM caller with retry and JSON extraction."""
    for attempt in range(max_retries):
        try:
            resp = client.chat.completions.create(
                model=VISION_MODEL,
                messages=[{'role':'system','content':system_prompt},
                          {'role':'user',  'content':content}],
                max_tokens=max_tokens, temperature=0.1
            )
            raw = resp.choices[0].message.content.strip()
            raw = re.sub(r'^```(?:json)?\s*', '', raw, flags=re.MULTILINE)
            raw = re.sub(r'\s*```$', '', raw, flags=re.MULTILINE)
            match = re.search(r'\{.*\}', raw, re.DOTALL)
            if match:
                parsed = json.loads(match.group())
                if 'purchase_probability' in parsed:
                    parsed['purchase_probability'] = max(0.0, min(1.0,
                        float(parsed['purchase_probability'])))
                return parsed
        except Exception as e:
            time.sleep(2**attempt)
    return {'purchase_probability': 0.5, 'confidence': 'low', 'key_reason': 'Ex

def build_content(title, price, image_url, prompt_template):
    img_b64, mime = fetch_b64(image_url)
    text = prompt_template.format(title=title, price=price)
    if img_b64:
        return [{'type':'image_url','image_url':{'url':f'data:{mime}; base64,{im
                {'type':'text','text':text}]
    return [{'type':'text','text':text}]

# ── Run Zero-Shot ────────────────────────────────────────────────
print('Running Zero-Shot predictions...')
zs_results = []
```

```
for i, row in df_test.iterrows():
    print(f'  [{i+1}/5] {str(row["title"])[:55]}...', end=' ')
    content = build_content(str(row['title']), float(row['price']),
                            str(row['image']), ZS_PROMPT_TEMPLATE)
    res = call_llm(AGENT_SYSTEM_PROMPT, content)
    res['title'] = str(row['title']); res['price'] = float(row['price'])
    zs_results.append(res)
    print(f"p={res['purchase_probability']:.3f} [{res['confidence']}]")
    time.sleep(REQUEST_DELAY)

print('Zero-Shot complete.')
```

```
Running Zero-Shot predictions...
  [1/5] Men's Vintage Premium Washed Shirt Navy Blue / L... p=0.400 [medium]
  [2/5] Ribbed Textured Knit Polo Shirt (US Only) Khaki / L... p=0.400 [medium]
  [3/5] Men's Rose Print Vintage Casual Button Short Sleeve Shi... p=0.400 [medi
  [4/5] Classic Casual Button Down Cotton Linen Shirt (Us Only)... p=0.400 [medi
  [5/5] Men's Casual Geo Pattern Short Sleeve Shirt & Shorts 2P... p=0.400 [medi
Zero-Shot complete.
```

Step 5: Random Few-Shot Prediction

4 random examples (2 ordered, 2 not-ordered) from the training set are shown to the LLM as reference before each evaluation.

```
# ── Random Few-Shot ──────────────────────────────────────────────────
ordered_ex     = df_history[df_history['ordered'] == 1].sample(2, random_state=
not_ordered_ex = df_history[df_history['ordered'] == 0].sample(2, random_state=
random_examples = pd.concat([ordered_ex, not_ordered_ex])

def build_few_shot_system(examples_df):
    text = 'You are an experienced online fashion buyer.\nPreviously evaluated
    for _, row in examples_df.iterrows():
        label = 'ORDERED (prob ≈ 0.8)' if row['ordered'] == 1 else 'NOT ORDERED
        text += f'Product: "{str(row["title"])[:80]}"  |  ${float(row["price"])
    text += 'Using these as reference, evaluate new products. Respond ONLY with
    return text

FS_RANDOM_SYSTEM = build_few_shot_system(random_examples)

FS_USER_TEMPLATE = '''Evaluate this product:
Title: "{title}"
Price: ${price:.2f}

Return ONLY:
{{
```

```
  "purchase_probability": <float 0.0-1.0>,
  "confidence": <"low"|"medium"|"high">,
  "key_reason": <one concise sentence>
}}'''

print('Running Random Few-Shot predictions...')
fs_random_results = []
for i, row in df_test.iterrows():
    print(f'  [{i+1}/5] {str(row["title"])[:55]}...', end=' ')
    content = build_content(str(row['title']), float(row['price']),
                            str(row['image']), FS_USER_TEMPLATE)
    res = call_llm(FS_RANDOM_SYSTEM, content)
    res['title'] = str(row['title']); res['price'] = float(row['price'])
    fs_random_results.append(res)
    print(f"p={res['purchase_probability']:.3f} [{res['confidence']}]")
    time.sleep(REQUEST_DELAY)

print('Random Few-Shot complete.')
```

```
Running Random Few-Shot predictions...
  [1/5] Men's Vintage Premium Washed Shirt Navy Blue / L... p=0.600 [medium]
  [2/5] Ribbed Textured Knit Polo Shirt (US Only) Khaki / L... p=0.600 [medium]
  [3/5] Men's Rose Print Vintage Casual Button Short Sleeve Shi... p=0.200 [low]
  [4/5] Classic Casual Button Down Cotton Linen Shirt (Us Only)... p=0.200 [medi
  [5/5] Men's Casual Geo Pattern Short Sleeve Shirt & Shorts 2P... p=0.200 [low]
Random Few-Shot complete.
```

Step 6: RAG-Enhanced Few-Shot Prediction

For each test product, the *k* most similar training products are retrieved via cosine similarity on pre-computed title embeddings. These become the few-shot examples — giving the LLM contextually relevant reference products instead of random ones. This allows the full training dataset to contribute to predictions without exceeding the context window.

```
# ── RAG Retrieval Function ──────────────────────────────────────────────
from sklearn.feature_extraction.text import TfidfVectorizer

K_NEIGHBOURS = 6   # retrieve 3 ordered + 3 not-ordered (balanced)

def retrieve_similar_examples(test_title, k=K_NEIGHBOURS):
    """
    Retrieve k/2 most similar ordered and k/2 most similar not-ordered
    training products by cosine similarity on title TF-IDF vectors.
    Falls back to pre-computed embeddings if available.
```

```
    """
    # Use TF-IDF on titles as a lightweight retrieval mechanism
    # (embeddings already parsed; TF-IDF is fast and interpretable)
    all_titles = df_history['title'].tolist() + [test_title]
    tfidf = TfidfVectorizer(ngram_range=(1,2), min_df=1)
    tfidf_mat = tfidf.fit_transform(all_titles)
    query_vec = tfidf_mat[-1]          # last row = test product
    train_vecs = tfidf_mat[:-1]        # all training products
    sims = cosine_similarity(query_vec, train_vecs).flatten()

    # Retrieve top-k/2 from each class separately for balance
    half_k = k // 2
    ordered_idx     = np.where(df_history['ordered'] == 1)[0]
    not_ordered_idx = np.where(df_history['ordered'] == 0)[0]

    top_ordered     = ordered_idx[np.argsort(sims[ordered_idx])[::-1][:half_k]]
    top_not_ordered = not_ordered_idx[np.argsort(sims[not_ordered_idx])[::-1][:

    retrieved_idx = np.concatenate([top_ordered, top_not_ordered])
    return df_history.iloc[retrieved_idx], sims[retrieved_idx]


def build_rag_system_prompt(examples_df, similarities):
    text = (
        'You are an experienced online fashion buyer.\n'
        'The following are the most similar products to the one you will evalua
        'along with their actual purchase outcomes from our sales history:\n\n'
    )
    for (_, row), sim in zip(examples_df.iterrows(), similarities):
        label = 'ORDERED (purchase prob ≈ 0.8)' if row['ordered'] == 1 \
                else 'NOT ORDERED (purchase prob ≈ 0.2)'
        text += (
            f'Similar product (similarity={sim:.2f}): "{str(row["title"])[:80]}
            f'Outcome: {label}\n\n'
        )
    text += (
        'Use these similar historical products to calibrate your probability es
        'Respond ONLY with valid JSON — no markdown, no commentary.'
    )
    return text


COT_USER_TEMPLATE = '''Evaluate this product for purchase probability.
Title: "{title}"
Price: ${price:.2f}

Think step by step:
1. Visual Appeal — how attractive is the product image? (1-10)
2. Price-Value — is the price competitive for this type?
3. Trend Fit — does it align with current fashion trends? (1-10)
4. Target Market — how broad is the potential buyer pool?
```

```
5. Similarity to Retrieved Products — does outcome pattern from similar product

Return ONLY:
{{
  "visual_appeal_score":    <integer 1-10>,
  "price_value_aligned":    <true or false>,
  "trend_fit_score":        <integer 1-10>,
  "target_market_breadth":  <"narrow"|"moderate"|"broad">,
  "rag_signal":             <"positive"|"neutral"|"negative">,
  "reasoning_summary":      <2-3 sentence narrative>,
  "purchase_probability":   <float 0.0-1.0>,
  "confidence":             <"low"|"medium"|"high">
}}'''

print('RAG retrieval functions ready.')
print(f'Will retrieve {K_NEIGHBOURS} examples per test product ({K_NEIGHBOURS//
```

```
RAG retrieval functions ready.
Will retrieve 6 examples per test product (3 ordered + 3 not-ordered)
```

```
# ── Run RAG-Enhanced CoT Prediction ──────────────────────────────────
print('Running RAG-Enhanced CoT predictions...')
print('(Retrieves most similar training products per query)\n')

rag_results = []
for i, row in df_test.iterrows():
    title = str(row['title'])
    price = float(row['price'])
    print(f'  [{i+1}/5] {title[:55]}...')

    # Retrieve similar training examples for this specific product
    examples, sims = retrieve_similar_examples(title, k=K_NEIGHBOURS)
    print(f'    Retrieved examples (similarity scores): {[round(s,2) for s in s

    # Build RAG system prompt with retrieved examples
    rag_system = build_rag_system_prompt(examples, sims)

    # Build CoT user content with image
    content = build_content(title, price, str(row['image']), COT_USER_TEMPLATE)

    res = call_llm(rag_system, content, max_tokens=400)
    res['title'] = title
    res['price'] = price
    res['retrieved_examples'] = examples['title'].tolist()
    res['retrieved_outcomes'] = examples['ordered'].tolist()
    rag_results.append(res)

    print(f"    p={res['purchase_probability']:.3f} [{res.get('confidence','?')
          f"RAG signal: {res.get('rag_signal','?')}")
    time.sleep(REQUEST_DELAY)
```

```
print('\nRAG-Enhanced CoT complete.')
```

```
Running RAG-Enhanced CoT predictions...
(Retrieves most similar training products per query)

  [1/5] Men's Vintage Premium Washed Shirt Navy Blue / L...
    Retrieved examples (similarity scores): [np.float64(0.69), np.float64(0.69),
    p=0.800 [high]  RAG signal: positive
  [2/5] Ribbed Textured Knit Polo Shirt (US Only) Khaki / L...
    Retrieved examples (similarity scores): [np.float64(0.85), np.float64(0.22),
    p=0.720 [medium]  RAG signal: positive
  [3/5] Men's Rose Print Vintage Casual Button Short Sleeve Shi...
    Retrieved examples (similarity scores): [np.float64(0.23), np.float64(0.23),
    p=0.700 [medium]  RAG signal: positive
  [4/5] Classic Casual Button Down Cotton Linen Shirt (Us Only)...
    Retrieved examples (similarity scores): [np.float64(0.78), np.float64(0.28),
    p=0.720 [medium]  RAG signal: positive
  [5/5] Men's Casual Geo Pattern Short Sleeve Shirt & Shorts 2P...
    Retrieved examples (similarity scores): [np.float64(0.13), np.float64(0.13),
    p=0.650 [medium]  RAG signal: positive

RAG-Enhanced CoT complete.
```

Step 7: Priority Ranking and Strategy Comparison

```
# ── Build Priority Ranking ──────────────────────────────────────────
titles = df_test['title'].tolist()

priority = pd.DataFrame({
    'title':         titles,
    'price':         df_test['price'].values,
    'ZeroShot_prob': [r['purchase_probability'] for r in zs_results],
    'FewShot_prob':  [r['purchase_probability'] for r in fs_random_results],
    'RAG_CoT_prob':  [r['purchase_probability'] for r in rag_results],
    'RAG_signal':    [r.get('rag_signal','neutral') for r in rag_results],
    'RAG_reason':    [r.get('reasoning_summary', r.get('key_reason','')) for r
    'confidence':    [r.get('confidence','medium') for r in rag_results],
})

# Weighted ensemble: RAG-CoT 60%, Few-Shot 25%, Zero-Shot 15%
# RAG-CoT gets highest weight — it uses the most training information
priority['LLM_Ensemble_prob'] = (
    priority['ZeroShot_prob'] * 0.15 +
    priority['FewShot_prob']  * 0.25 +
    priority['RAG_CoT_prob']  * 0.60
).round(4)
```

```
priority['RAG_Rank']      = priority['RAG_CoT_prob'].rank(ascending=False).asty
priority['Ensemble_Rank'] = priority['LLM_Ensemble_prob'].rank(ascending=False)

print('=== LLM PRODUCT PRIORITY RANKING ===')
pd.set_option('display.max_colwidth', 50)
print(priority.sort_values('RAG_Rank')[
    ['RAG_Rank','Ensemble_Rank','title','price',
     'ZeroShot_prob','FewShot_prob','RAG_CoT_prob','LLM_Ensemble_prob']
].to_string(index=False))
```

```
=== LLM PRODUCT PRIORITY RANKING ===
 RAG_Rank  Ensemble_Rank
        1             1                           Men's Vintage Premium Washed
        2             2                       Ribbed Textured Knit Polo Shirt
        2             3            Classic Casual Button Down Cotton Linen Shir
        4             4        Men's Rose Print Vintage Casual Button Short Sle
        5             5 Men's Casual Geo Pattern Short Sleeve Shirt & Shorts 2P
```

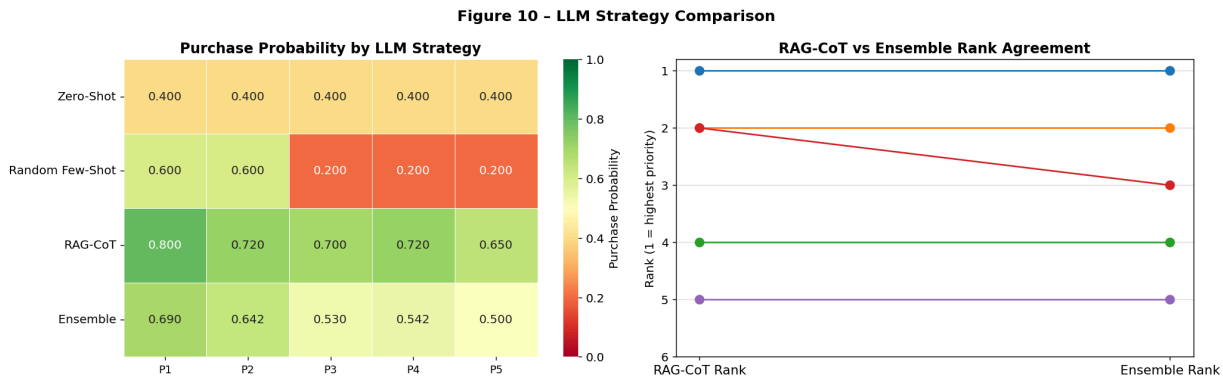
```
# ── Figure 10: Strategy Comparison Heatmap ──────────────────────────
fig, axes = plt.subplots(1, 2, figsize=(16, 5))

heat_data = priority[['ZeroShot_prob','FewShot_prob','RAG_CoT_prob','LLM_Ensemb
heat_data.columns = [f'P{i+1}' for i in range(len(priority))]
heat_data.index   = ['Zero-Shot','Random Few-Shot','RAG-CoT','Ensemble']

sns.heatmap(heat_data, annot=True, fmt='.3f', cmap='RdYlGn',
            vmin=0, vmax=1, ax=axes[0], linewidths=0.5,
            cbar_kws={'label':'Purchase Probability'})
axes[0].set_title('Purchase Probability by LLM Strategy', fontweight='bold')
axes[0].set_xticklabels([f'P{i+1}' for i in range(len(priority))], fontsize=10)

# RAG vs Random rank agreement
for _, row in priority.iterrows():
    axes[1].plot([1,2], [row['RAG_Rank'], row['Ensemble_Rank']],
                 'o-', markersize=8, linewidth=1.5)
axes[1].set_xticks([1,2])
axes[1].set_xticklabels(['RAG-CoT Rank','Ensemble Rank'], fontsize=12)
axes[1].set_yticks(range(1, len(priority)+2))
axes[1].invert_yaxis()
axes[1].set_ylabel('Rank (1 = highest priority)', fontsize=11)
axes[1].set_title('RAG-CoT vs Ensemble Rank Agreement', fontweight='bold')
axes[1].grid(axis='y', alpha=0.4)

plt.suptitle('Figure 10 – LLM Strategy Comparison', fontsize=14, fontweight='bo
plt.tight_layout()
plt.savefig('fig10_llm_comparison.png', dpi=150, bbox_inches='tight')
plt.show()
```



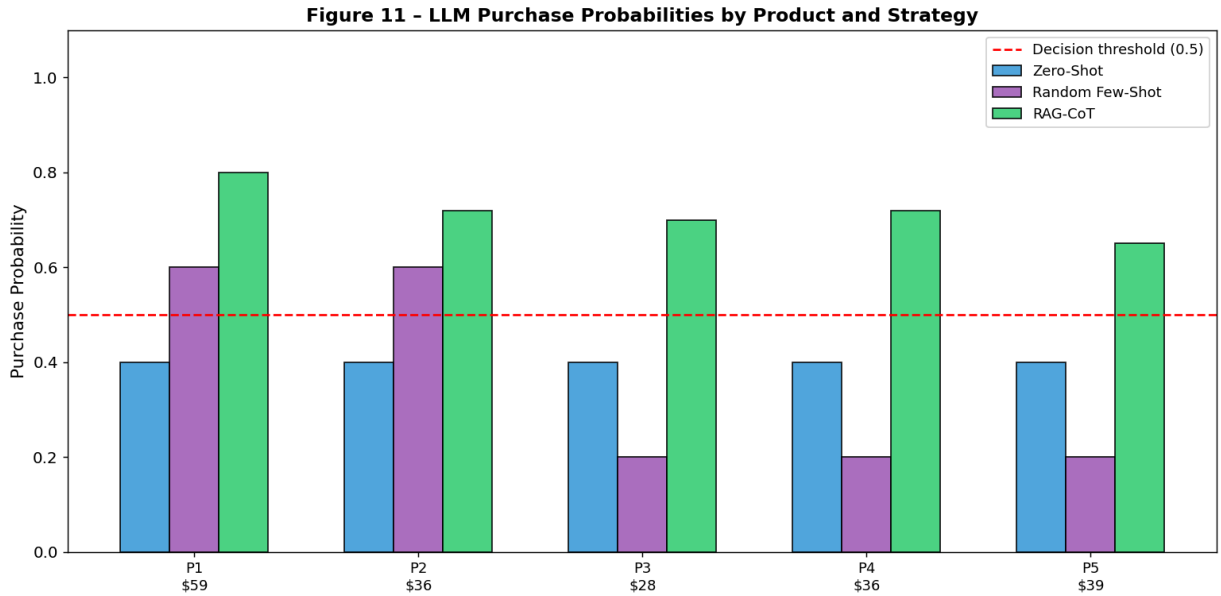

```
# ── Figure 11: Final Priority Ranking Bar Chart ──────────────────────
fig, ax = plt.subplots(figsize=(12, 6))

x     = np.arange(len(priority))
width = 0.22

ax.bar(x - 1.0*width, priority['ZeroShot_prob'],  width, label='Zero-Shot',
ax.bar(x + 0.0*width, priority['FewShot_prob'],   width, label='Random Few-Shot
ax.bar(x + 1.0*width, priority['RAG_CoT_prob'],   width, label='RAG-CoT',

ax.axhline(0.5, color='red', linestyle='--', lw=1.5, label='Decision threshold
ax.set_xticks(x)
ax.set_xticklabels(
    [f'P{i+1}\n${priority.iloc[i]["price"]:.0f}' for i in range(len(priority))]
    fontsize=10
)
ax.set_ylabel('Purchase Probability', fontsize=12)
ax.set_ylim(0, 1.1)
ax.set_title('Figure 11 – LLM Purchase Probabilities by Product and Strategy',
             fontsize=13, fontweight='bold')
ax.legend(fontsize=10)
plt.tight_layout()
plt.savefig('fig11_llm_ranking.png', dpi=150, bbox_inches='tight')
plt.show()

print('\nProduct Legend:')
for i, t in enumerate(priority['title']):
    print(f'  P{i+1}: {t[:80]}')
```




```
Product Legend:
  P1: Men's Vintage Premium Washed Shirt Navy Blue / L
  P2: Ribbed Textured Knit Polo Shirt (US Only) Khaki / L
  P3: Men's Rose Print Vintage Casual Button Short Sleeve Shirt WHITE / L
  P4: Classic Casual Button Down Cotton Linen Shirt (Us Only) Red / L
  P5: Men's Casual Geo Pattern Short Sleeve Shirt & Shorts 2Pcs Set Burgundy / L
```

```
# ── RAG Reasoning Summaries ──────────────────────────────────────────────
print('\n=== RAG-CoT AGENT REASONING SUMMARIES ===')
for _, row in priority.sort_values('RAG_Rank').iterrows():
    print(f"\nRank #{row['RAG_Rank']}: {str(row['title'])[:70]}")
    print(f"  RAG-CoT Prob  : {row['RAG_CoT_prob']:.3f}  |  Confidence: {row['c
    print(f"  RAG Signal    : {row['RAG_signal']}")
    print(f"  Ensemble Prob : {row['LLM_Ensemble_prob']:.3f}")
    print(f"  Reasoning     : {row['RAG_reason']}")

    # Show what was retrieved
    idx = priority[priority['title']==row['title']].index[0]
    retrieved = rag_results[idx]
    print(f"  Retrieved examples:")
    for t, o in zip(retrieved.get('retrieved_examples',[]), retrieved.get('retr
        label = '✓ ordered' if o==1 else '✗ not ordered'
        print(f"    [{label}] {t[:60]}")
```

```
    Ensemble Prob : 0.642
    Reasoning     : The product image is visually appealing with a clear view of
    Retrieved examples:
      [✓ ordered] Ribbed Textured Knit Polo Shirt (US Only) Pink / M
      [✓ ordered] Men's Cotton Knit Striped Polo Shirt Khaki / 4XL
      [✓ ordered] Men's Cotton Knit Striped Polo Shirt Purple / M
      [✗ not ordered] Cotton Linen Style Henley Shirt (US Only) Khaki / L
      [✗ not ordered] Classic Casual Button Down Cotton Linen Shirt (Us Only) Kh
      [✗ not ordered] Casual Loose Fit Shirt Sets (US Only) Khaki / M

  Rank #2: Classic Casual Button Down Cotton Linen Shirt (Us Only) Red / L
    RAG-CoT Prob  : 0.720  |  Confidence: medium
    RAG Signal    : positive
    Ensemble Prob : 0.542
    Reasoning     : The product has a high visual appeal and aligns well with cu
    Retrieved examples:
      [✓ ordered] Classic Casual Button Down Cotton Linen Shirt (Us Only) Wine
      [✓ ordered] Embroidered Cowboy Button Down Shirt (US Only) Classic Black
      [✓ ordered] Casual Cotton Linen Shirt Set White / M
      [✗ not ordered] Classic Casual Button Down Cotton Linen Shirt (Us Only) Kh
      [✗ not ordered] Classic Casual Button Down Cotton Linen Shirt (Us Only) Pu
      [✗ not ordered] Classic Casual Button Down Cotton Linen Shirt (Us Only) Pu

  Rank #4: Men's Rose Print Vintage Casual Button Short Sleeve Shirt WHITE / L
    RAG-CoT Prob  : 0.700  |  Confidence: medium
    RAG Signal    : positive
    Ensemble Prob : 0.530
    Reasoning     : The product has a unique rose print design which could appea
    Retrieved examples:
      [✓ ordered] Men's Linen Resort Casual Button-Down Everyday Short Sleeve
      [✓ ordered] Men's Linen Resort Casual Button-Down Everyday Short Sleeve
      [✓ ordered] Men's Linen Resort Casual Button-Down Everyday Short Sleeve
      [✗ not ordered] Men's Linen Resort Casual Button-Down Everyday Short Sleev
      [✗ not ordered] Men's Linen Resort Casual Button-Down Everyday Short Sleev
      [✗ not ordered] Men's Linen Resort Casual Button-Down Everyday Short Sleev

  Rank #5: Men's Casual Geo Pattern Short Sleeve Shirt & Shorts 2Pcs Set Burgundy
    RAG-CoT Prob  : 0.650  |  Confidence: medium
    RAG Signal    : positive
    Ensemble Prob : 0.500
    Reasoning     : The product has a visually appealing geometric pattern and a
    Retrieved examples:
      [✓ ordered] Men's Linen Resort Casual Button-Down Everyday Short Sleeve
      [✓ ordered] Men's Linen Resort Casual Button-Down Everyday Short Sleeve
      [✓ ordered] Men's Linen Resort Casual Button-Down Everyday Short Sleeve
      [✗ not ordered] Men's Casual Corduroy Pocket Short Sleeve Shirt Black / L
      [✗ not ordered] Men's Casual Corduroy Pocket Short Sleeve Shirt Brown / L
      [✗ not ordered] Minimal Comfort Polo Shirt Shorts Set White / L
```

```
# ── Export ──────────────────────────────────────────────────────────
priority.to_csv('llm_priority_ranking.csv', index=False)
from google.colab import files
files.download('llm_priority_ranking.csv')

print('\n✅ Part D Complete.')
```

```
print('\nFinal RAG-CoT Priority Ranking:')
for _, row in priority.sort_values('RAG_Rank').iterrows():
    bar = '└' * int(row['RAG_CoT_prob']*20)
    print(f"  #{row['RAG_Rank']} [{row['RAG_CoT_prob']:.3f}] {bar} {str(row
```

```
✅ Part D Complete.

Final RAG-CoT Priority Ranking:
  #1 [0.800] └└└└└└└└└└└└└└└└└└└└ Men's Vintage Premium Washed Shirt Navy Blue / L
  #2 [0.720] └└└└└└└└└└└└└└└└└└ Ribbed Textured Knit Polo Shirt (US Only) Khaki / L
```